WASP-43b

R-0594

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_230223` · created 2026-07-24T23:56:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459998.9058 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.058
Transit depth (Rp/Rs)^2	3.67 +/- 2.22 [%]
Orbital Inclination (inc)	86.96 +/- 2.93
Airmass coefficient 1 (a1)	0.59 +/- 0.22
Airmass coefficient 2 (a2)	0.28 +/- 0.23
Scatter in the residuals of the lightcurve fit is	37.14 %
Best Comparison Star	#3 - [577, 310]
Optimal Aperture	11.14
Optimal Annulus	11.59
Transit Duration (day)	0.0612 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.058 vs 0.1594 (deviation 0.54σ)
Duration fit vs published	0.0612 d vs 0.0512 d (deviation 2.07σ)
Mid-time O–C	-4.2 min
Depth SNR	3.3
Residual scatter	37.14 %

Light curve

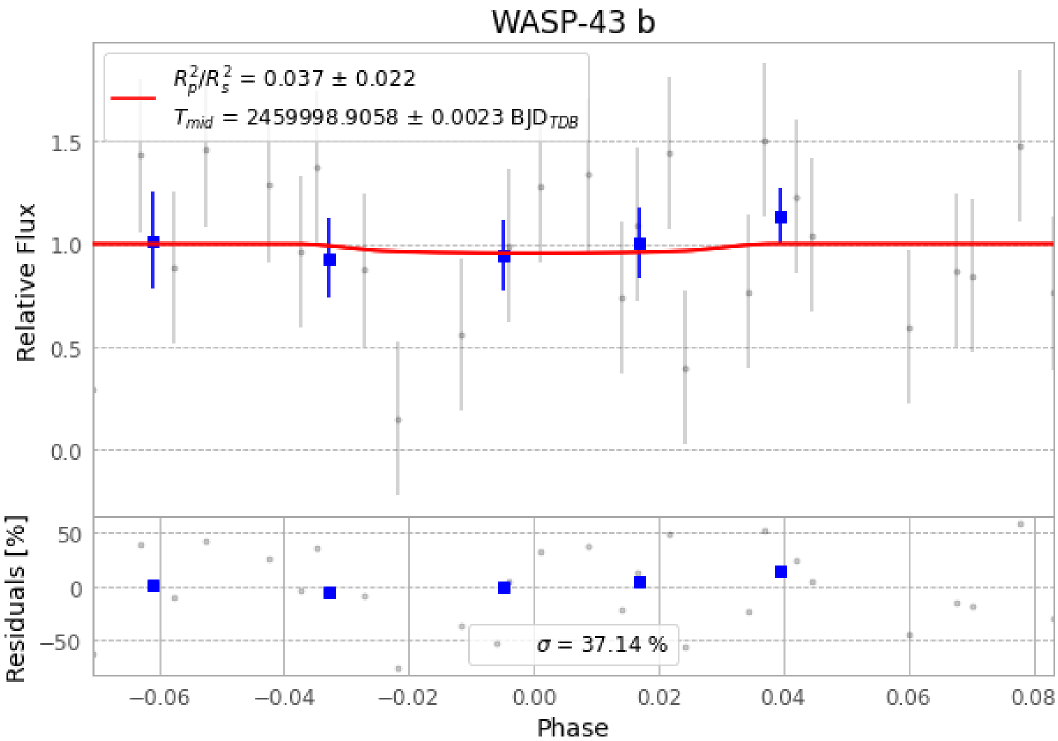

Figure 1 — FinalLightCurve_WASP-43 b_2023-02-23.png (SHA-256 e6f7518bbd43a9ab…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [497, 103], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3b733a02611f830e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

57 FITS frames (37 MB), `WASP-43230223080915.FITS` → `WASP-43230223111515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230223.log (64dff9c265ec…), FinalLightCurve_WASP-43 b_2023-02-23.png (e6f7518bbd43…), FinalLightCurve_WASP-43 b_2023-02-23.csv (c580c7572af0…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)